

Is Mrs. Gallas a Good Free Throw Shooter?



Mrs. Gallas was a star basketball player in high school, where she was an impressive 70% free throw shooter.

1. In your own words, what does it mean to be a 70% free throw shooter?

2. If Mrs. Gallas takes 100 shots, how many do you expect she will make? Is this guaranteed?

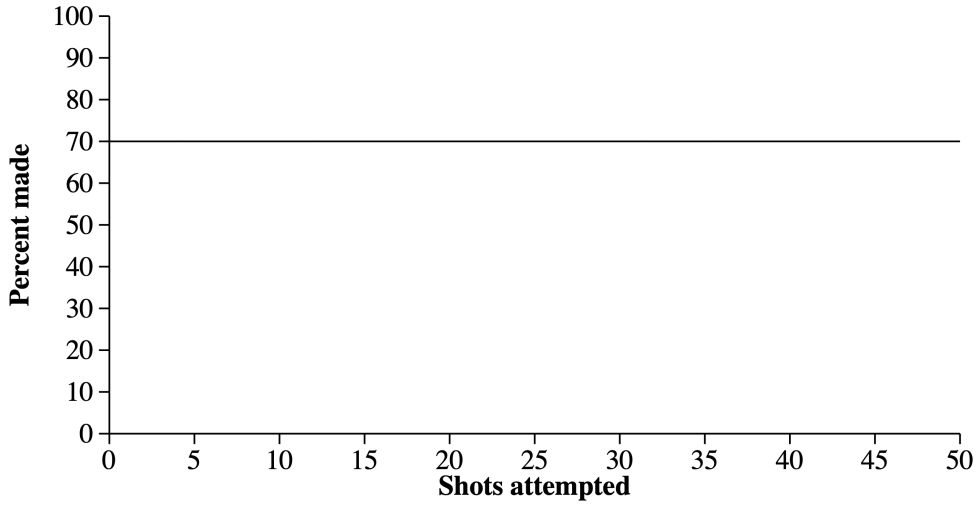
3. If Mrs. Gallas takes 100 shots, do you think she will have a streak where she makes 10 in a row?

4. Go to stapplet.com/largenum.html. Use the applet to shoot 100 free throws and use the table to keep track of the results.

Shot #	1	2	3	4	5	10	15	20	30	40	50	100
Result (make or miss)												
Percent made												

5. Sketch the graph displaying the percent made after each shot attempted.

6. What do you notice in the graph?



7. In the 100 shots, did Mrs. Gallas make exactly 70 free throws? Why or why not?

8. Based on the results of the applet, what does a probability of 70% mean?

Lesson 3.1 – Introducing Probability

QuickNotes

Check Your Understanding

1. Choose a probability value to complete the middle column of the table, then give an example of an event that may have this probability. Probabilities: 0, 0.001, 0.3, 0.6, 0.99, 1

Explanation	Probability	Example
This outcome is impossible. It can never occur.		
This outcome is certain. It will occur on every trial.		
This outcome is very unlikely, but it will occur once in a while in a long sequence of trials.		
This outcome will occur more often than not, but doesn't occur almost every time.		

2. In the Wheel of Fortune, there is a 1/9 = 11.1% probability of spinning "Bankrupt" on any given spin. Interpret the probability.

3. In 9 consecutive spins of the Wheel of Fortune, none of the spins land on "Bankrupt". The next contestant worries that he is more likely to land on Bankrupt now because Bankrupt is due. Explain why this thinking is wrong.